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Employment quality and mortality in Canada

Faraz V Shahidi ,[1,2] Alessandra T Andreacchi,[1,2] Anne E Fuller ,[3,4] Alexandra Blair,[5] Nancy Carnide ,[1,2] M Anne Harris,[6] Roman Pabayo ,[7] Brendan Smith ,[8] Arjumand Siddiqi ,[9] Peter Smith [1,9]

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[1]Institute for Work and Health, Toronto, Ontario, Canada
[2]University of Toronto, Toronto, Ontario, Canada
[3]Department of Pediatrics, McMaster University Faculty of Health Sciences, Hamilton, Ontario, Canada
[4]Pediatrics, McMaster Children's Hospital, Hamilton, Ontario, Canada
[5]Public Health Agency of Canada, Ottawa, Ontario, Canada
[6]Toronto Metropolitan University, Toronto, Ontario, Canada
[7]School of Public Health, University of Alberta, Edmonton, Alberta, Canada
[8]Public Health Ontario, Toronto, Ontario, Canada
[9]Dalla Lana School of Public Health, University of Toronto, Toronto, Ontario, Canada

Correspondence to
Dr Faraz V Shahidi; fshahidi@iwh.on.ca

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ABSTRACT

Background Research has shown that workers in non-standard (eg, temporary and part-time) employment experience poorer health outcomes than their permanent, full-time counterparts. However, previous studies have overlooked important differences in the quality of non-standard employment. To address this gap, we examined associations between a diverse typology of employment quality and mortality in Canada.

Methods The 2006 Canadian Health and Environment Cohort (n=2 805 550) was linked to death records from 2006 to 2019. Employment quality was assessed according to an empirical typology describing five distinct employment arrangements: standard (secure and gainful), portfolio (demanding but gainful), marginal (limited hours and earnings), intermittent (sporadic and unstable) and precarious (insecure and low paying). Poisson regression models estimated covariate-adjusted associations between employment quality, all-cause and cause-specific (cancer, cardiovascular and unintentional injury) mortality, by sex/gender.

Results We observed a graded association between employment quality and mortality. Mortality rates were lowest among workers in standard and portfolio employment. Mortality rates were highest among workers in precarious employment, with workers in marginal and intermittent employment occupying intermediate positions along the risk gradient. Associations varied by sex/gender, with larger absolute and relative mortality inequalities among men.

Conclusions Our findings reinforce the need to move away from a binary view of jobs as either 'standard' or 'precarious', encouraging a more nuanced understanding of contemporary employment arrangements and their health-related consequences. Policy interventions that promote access to high-quality jobs and protect workers exposed to precarious employment may yield substantial improvements in population health, including longevity.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Non-standard employment is associated with adverse health outcomes, including poorer general and mental health.
⇒ Substantial differences exist in the quality of non-standard employment—while some non-standard jobs are precarious, others are not.
⇒ We lack a nuanced understanding of how different types of non-standard employment associate with health outcomes in general, and mortality in particular.

WHAT THIS STUDY ADDS

⇒ This study examined the relationship between a diverse typology of employment and all-cause and cause-specific (cancer, cardiovascular and unintentional injury) mortality in Canada.
⇒ We identified five distinct employment arrangements of varying quality: standard, portfolio, marginal, intermittent and precarious.
⇒ Mortality rates were lowest among workers in standard and portfolio employment, and highest among workers in precarious employment.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ A nuanced view of contemporary employment arrangements is important for understanding the health consequences of non-standard and precarious employment.
⇒ Policy interventions that promote access to high-quality jobs and protect workers exposed to precarious employment may yield substantial improvements in population health, including longevity.

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INTRODUCTION

Social and economic forces such as neoliberalism, globalisation and deindustrialisation have driven profound changes in the nature of work and contributed to an increasingly precarious employment landscape in Canada and other high-income countries.[1 2] These labour market trends are often understood in terms of a long-run erosion of the so-called 'Standard Employment Relationship' (typically defined as permanent, full-time employment with benefits), with a corresponding increase in the prevalence of temporary, part-time and other non-standard employment arrangements.[3–5] The flexibility afforded by some non-standard jobs is thought to benefit certain groups, such as parents and caregivers who balance competing work and non-work demands.[6] For most workers, however, the decline of the 'Standard Employment Relationship' has instead implied heightened levels of precarity in the labour market, with adverse consequences for health and well-being.[1–3 7] This redistribution of risk in the labour market is both a symptom of and a driving force behind the growing power imbalance between employers and workers.[8]

Previous literature has shown that workers in non-standard jobs experience poorer health outcomes than their permanent, full-time counterparts.[3 9] However, evidence to that effect is largely

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limited to subjective measures of general and mental health.[9–15] For example, there is an extensive body of research linking non-standard employment to self-reported symptoms of depression and psychological distress.[12–15] By contrast, evidence on clinically significant end points such as mortality is lacking.[8 16] While recent studies suggest that non-standard employment is associated with an increased risk of death, they are limited in number.[16–20] More importantly, these studies typically rely on a dichotomous view of jobs as either 'standard' or 'precarious'.[16] Yet not all jobs that deviate from the 'standard' model of employment are necessarily precarious in nature.[21] The quality of non-standard employment can instead vary across multiple dimensions of work (eg, hours, stability and earnings).[22] Collapsing these heterogeneous arrangements into a single 'precarious' category may obscure important differences in the nature and character of non-standard employment, resulting in a biased picture of the employment-health relationship.[23 24]

At present, we lack a nuanced understanding of how different types of non-standard employment influence health outcomes in general and mortality in particular. This study addresses that gap by examining the relationship between a diverse typology of employment and both all-cause and cause-specific (cancer, cardiovascular and unintentional injury) mortality in Canada, with the aim of providing a more nuanced view of contemporary employment arrangements and their health-related consequences.

METHODS

Data and sample

Data were drawn from the 2006 Canadian Census Health and Environment Cohort (CanCHEC), a nationally representative cohort of nearly 6 million Canadians selected to complete the mandatory long-form Census in May 2006.[25] Census records were linked to the Canadian Vital Statistics Death Database (CVSD), containing a detailed census of deaths in Canada.[25] Record linkage was completed by Statistics Canada and is available for deaths recorded up to 31 December 2019. Deaths recorded beyond that date are not presently available for linkage to the 2006 CanCHEC. The response rate to the long-form Census was 93.8%. Approximately 90.8% of long-form Census respondents were successfully linked to the CVSD. Response and linkage rates differed by characteristics, including age, marital status, race/ethnicity, education and income.[25] Statistics Canada provides survey weights to account for resulting sampling bias.

Our population of interest consisted of adults who were 18–64 years of age and active in the labour force (ie, either employed or unemployed and actively seeking work). From an initial sample of 3 034 625 CanCHEC respondents who met these criteria, we excluded 122 375 respondents with no record of employment in 2005, as well as 106 160 respondents who were temporarily absent from work and lacking key information on the exposure. We also excluded 540 respondents who were missing covariate information. Our final analytic sample consisted of 2 805 550 CanCHEC respondents with linked census and mortality records. A study flow diagram is available in figure 1. (All sample sizes are randomly rounded to the nearest five in accordance with Statistics Canada guidelines.)

Exposure: employment quality

Employment quality was assessed according to a previously constructed typology, details of which are presented elsewhere.[26] Following recent studies in Europe and the USA,[4 5 27] we used latent class analysis (LCA) to group workers into various 'segments' of the labour market, each characterised by a unique

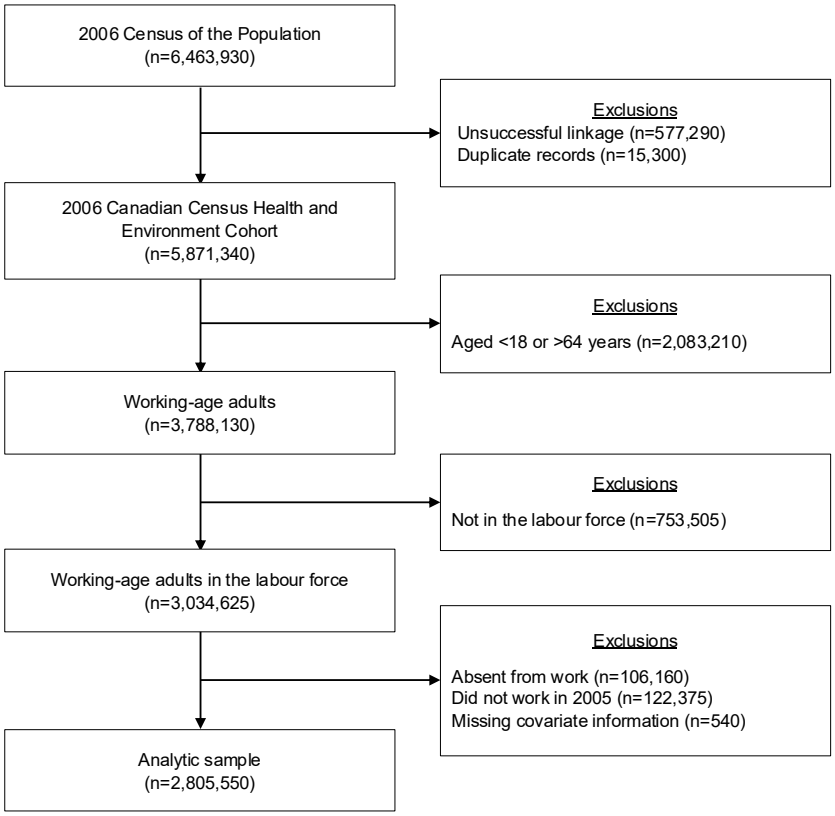

Figure 1 Study flow chart for exclusions from the 2006 Canadian Census Health and Environment Cohort (CanCHEC).

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Table 1 An overview of the latent typology of employment quality

Latent class	Key characteristics	Weighted proportion
Standard employment	Standard employment was characterised by secure and gainful employment conditions. Workers in this group were very likely to work standard full-time hours and maintain employment throughout the year. Their work was generally gainful, with above-average employment earnings. Examples observed include civil servants and other workers in the public sector, as well as private-sector employees in industries with a strong tradition of collective bargaining (eg, manufacturing and utilities).	44.0%
Portfolio employment	Portfolio employment was characterised by demanding but gainful employment conditions. Virtually all workers in this group reported working long hours (ie, 49 or more hours per week). However, their work was relatively stable, and they reported the highest levels of employment earnings. Examples observed include self-employed professionals (eg, doctors, lawyers and accountants), as well as independent contractors and senior managers in the private sector.	14.9%
Marginal employment	Marginal employment was characterised by a combination of limited hours and low earnings. Part-time and part-year employment were both very prevalent within this group, with a majority of its members working fewer than 30 hours per week and fewer than 26 weeks of the year. Examples observed include part-time and casual workers in the education and healthcare sectors (eg, substitute teachers, tutors, home health aides and part-time nurses).	13.2%
Intermittent employment	Intermittent employment was characterised by unstable and sporadic employment conditions. Workers in this group were very likely to work full-time hours. However, they reported substantial interruptions to employment throughout the year and below-average earnings as a result. Examples observed include seasonal workers in trades and construction (eg, roofers, loggers and road workers), as well as workers in industries such as hospitality and tourism, where demand for labour is subject to seasonal fluctuations.	16.3%
Precarious employment	Precarious employment was characterised by insecure and economically unrewarding employment conditions. Workers in this group were very likely to work non-standard hours. They were also very likely to work less than half of the year and reported the lowest levels of employment earnings. Examples observed include low-wage workers in retail and food services, as well as casual labourers in the construction and agricultural sectors.	11.6%

configuration of employment conditions. We included three indicators in the LCA: employment hours, employment stability and employment earnings. These indicators align with multiple dimensions of an established framework for the study of employment quality as a social determinant of health.[23 24] *Employment hours* refer to the number of hours worked during the previous week. Respondents were grouped into four categories according to relevant labour laws and the thresholds used by Canada's national statistical agency: 0 hours (ie, unemployed and looking for work); 1–14 hours; 15–29 hours; 30–48 hours and 49 or more hours. *Employment stability* refers to the number of weeks worked in the previous calendar year. Respondents were grouped into three categories: 1–25 weeks (ie, less than half the year); 26–51 weeks (ie, at least half the year but not the full year) and 52 weeks (ie, the full year). (3) *Employment earnings* refer to gross personal income from wages, salaries or tips during the previous calendar year. Respondents were grouped into five quintiles ranging from Q1 (lowest) to Q5 (highest).

The LCA supported a five-class typology of employment quality.[26] We labelled the five groups, generally in order from highest to lowest employment quality, as follows: 'standard employment' (44.0%), 'portfolio employment' (14.9%), 'marginal employment' (13.2%), 'intermittent employment' (16.3%) and 'precarious employment' (11.6%). Key features of these groups, including illustrative examples observed, are described in table 1. Standard employment was characterised by a high probability of full-time, stable and gainful employment conditions that closely approximate the so-called 'Standard Employment Relationship'. Portfolio employment—a term we borrow from earlier studies[4 5]—was also characterised by favourable employment conditions, including the highest earnings. However, unlike their counterparts in standard employment, virtually all workers in the portfolio group reported working long hours. The 'portfolio' label is widely applied in the employment quality literature to describe employment arrangements that combine long hours with high rewards.[4 5 21–24] However, this group could also be described as an example of 'intensive' employment.[4] Marginal employment was characterised by a combination of limited hours and low earnings—features that suggest underemployment or participation at the margins (or periphery) of the labour market. Intermittent employment was characterised by a high probability of full-time hours, although coupled with a very low probability of year-round employment, suggesting a predominance of seasonal or irregular work. Finally, precarious employment was characterised by the least favourable employment conditions, with a high probability of non-standard working hours, a high degree of employment instability and the lowest earnings of all. While marginal and intermittent employment exhibited some unfavourable features, this precarious group is specifically characterised by multiple, overlapping experiences of adversity across all three of the employment quality dimensions.[7] This typology aligns in key respects with empirical rubrics constructed in Europe and the USA, which have characterised both the 'Standard Employment Relationship' and various forms of non-standard employment, including the 'portfolio' and 'precarious' groups identified here.[4 5] For additional information about these groups, see online supplemental appendix A.

Outcome: mortality

Mortality was assessed using all available death records from the CVSD. The study end points were death or censoring on the last available day of follow-up (ie, 31 December 2019). Underlying cause of death was determined using the International Classification of Diseases, 10th Revision (ICD-10) codes. In addition to all-cause mortality, we examined cancer mortality (ICD-10 C00-C97), cardiovascular mortality (ICD-10 I00-I99) and unintentional injury mortality (ICD-10 V01-X59, Y85-Y86). The latter outcomes were selected because they represent the top three leading causes of death in Canada. We were unable to identify loss to follow-up due to emigration. Emigrants were therefore treated as survivors.

Covariates

We collected information on sex/gender (man or woman), age group (18–24, 25–34, 35–44, 45–54 or 55–64 years), race/ethnicity (white, black, East/Southeast Asian, Latin American, Arab/West Asian, South Asian or other/multiple), immigrant status (born in Canada or born outside Canada), marital status

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(single, married/cohabitating or separated/divorced/widowed), household size (one, two, three or four or more), region of residence (Eastern Canada (Newfoundland and Labrador, Nova Scotia, Prince Edward Island, New Brunswick), Central Canada (Ontario, Quebec), Western Canada (British Columbia, Alberta, Saskatchewan, Manitoba) or Northern Canada (Yukon, Northwest Territories, Nunavut)), rurality (rural or urban) and education (less than high school, high school diploma, postsecondary below bachelor's degree, bachelor's degree or higher). Whereas the 2006 Census only measured sex, we recognise that this measure captures the influence of both sex and gender on health and labour market outcomes. We therefore interpreted this measure as an indicator of 'sex/gender'.

Statistical analysis

Analyses were stratified by sex/gender due to evidence that the relationship between employment quality and health can vary between women and men.[28] We calculated age-standardised mortality rates for each of the latent employment quality groups, using the direct method of standardisation and taking the 5-year age distribution of the 2011 Canadian Census Population, currently the most widely used reference population in Canada. Sex-stratified/gender-stratified Poisson regression models were then used to estimate rate ratios (RR) and corresponding 95% CIs describing the association between employment quality and mortality, with standard employment as the reference group. To account for varying length of follow-up and competing risks of death, we included log person-years as an offset. We estimated a single adjusted model in which posterior probabilities of membership in the latent classes were included as continuous variables alongside the full set of covariates, following the same approach as previous research in this area.[23] Sensitivity analyses employing Cox proportional hazards models produced equivalent results. We selected Poisson regression as our primary approach because it is computationally more efficient, produces more readily interpretable results for public health audiences and allows direct estimation of absolute rates and rate differences.

We conducted two additional sensitivity analyses. First, to address potential age-related differences in the meaning and consequences of employment quality, we stratified our analyses by age, distinguishing between younger workers (aged 18–44 years) and older workers (aged 45–64 years). Second, we excluded respondents who died within the first 5 years of follow-up, reducing the risk of unmeasured confounding due to unobserved variables associated with imminent death that also affect employment quality (eg, pre-existing health conditions).

We applied the aforementioned survey weights provided by Statistics Canada to ensure the representativeness of our findings.

RESULTS

The sociodemographic characteristics of the study population are presented in table 2. The sociodemographic characteristics of the latent employment quality groups are described in online supplemental appendix B. Compared with those in higher quality (ie, standard or portfolio) employment, workers in lower quality (ie, marginal, intermittent or precarious) employment were more likely to be younger, single, non-white and born outside of Canada. They were also less likely to have obtained a university degree. A greater proportion of women were in marginal and precarious employment, while men were more likely to be in portfolio employment.

Covariate-adjusted associations between the employment quality typology and all-cause mortality are presented in table 3, where we also describe the number of deaths, person-years and age-standardised mortality rates. Between May 2016 and December 2019, we observed a total of 514 855 deaths in the weighted sample representing 7 113 445 women and 8 067 890 men, with a mean follow-up time of 13.4 years. We found a graded association between employment quality and mortality. Mortality rates were lowest among workers in standard and portfolio employment and highest among those in precarious employment, with workers in intermittent and marginal employment occupying intermediate positions in the gradient. A similar overall pattern was observed among men and women alike, although with larger absolute rate differences among men due to their higher baseline rate of mortality. Compared with standard employment, portfolio employment did not meaningfully predict differences in mortality among either women (RR 1.09, 95% CI 1.06 to 1.11) or men (RR 0.98, 95% CI 0.97 to 0.99).

Table 2 Sociodemographic characteristics of the study population, by sex/gender: 2006 Canadian Census Health and Environment Cohort

	Women	Men
Weighted n	7 113 445	8 067 890
Age group (years), %		
18–24	14.2	13.4
25–34	20.4	21.1
35–44	26.1	25.6
45–54	26.8	25.7
55–64	12.4	14.2
Race/Ethnicity, %		
White	82	82.1
Black	2.3	2.1
East/Southeast Asian	6.7	5.9
Latin American	1.0	1.0
Arab/West Asian	0.9	1.3
South Asian	3.5	4.1
Other/Multiple	3.6	3.4
Immigration status, %		
Born in Canada	78.6	78.0
Born outside Canada	21.4	22.0
Marital status, %		
Single	25.6	27.5
Married or co-habiting	62.7	65.7
Divorced, separated or widowed	11.7	6.7
Household size, %		
1	9.9	10.8
2	29.7	26.4
3	21.4	21.4
4+	38.9	41.4
Region of residence, %		
Eastern Canada	7.1	6.9
Central Canada	62.2	62.0
Western Canada	30.4	30.8
Northern Canada	0.3	0.3
Rurality, %		
Rural	18.3	19.8
Urban	81.7	80.2
Education, %		
Less than high school	9.9	14.3
High school diploma	27.5	26.3
Some postsecondary	38.4	37.8
Bachelor's degree or higher	24.1	21.6

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Table 3 Estimates from adjusted Poisson regression models describing the relationship between employment quality and all-cause mortality by sex/gender: 2006 Canadian Census Health and Environment Cohort

	Deaths	Person-years	ASMR (95% CI)	RR* (95% CI)
Women				
Standard employment	76 485	41 572 610	283.1 (280.8 to 285.5)	Reference
Portfolio employment	14 460	7 703 000	281.5 (276.2 to 286.9)	1.09 (1.06 to 1.11)
Marginal employment	35 390	17 527 650	314.3 (310.9 to 317.6)	1.24 (1.22 to 1.26)
Intermittent employment	28 450	16 064 725	319.1 (314.9 to 323.3)	1.26 (1.23 to 1.29)
Precarious employment	21 825	13 124 490	371.2 (366.0 to 376.4)	1.50 (1.48 to 1.53)
Men				
Standard employment	146 245	48 171 360	444.7 (442.2 to 447.2)	Reference
Portfolio employment	66 380	22 704 880	421.1 (417.5 to 424.7)	0.98 (0.97 to 0.99)
Marginal employment	35 190	9 346 075	568.2 (562.3 to 574.1)	1.40 (1.38 to 1.42)
Intermittent employment	56 115	17 264 725	568.2 (563.4 to 572.9)	1.40 (1.38 to 1.43)
Precarious employment	34 315	10 605 195	683.3 (676.0 to 690.7)	1.60 (1.58 to 1.62)

*Adjusted for age, race/ethnicity, immigrant status, marital status, region of residence, rurality, education and household size.
ASMR, age-standardised mortality rate per 10 000 persons; RR, rate ratio.

Marginal and intermittent employment were associated with a roughly 25% higher mortality rate among women (RR 1.26, 95% CI 1.23 to 1.29 and RR 1.24, 95% CI 1.22 to 1.26, respectively) and a roughly 40% higher mortality rate among men (RR 1.40, 95% CI 1.38 to 1.43 and RR 1.40, 95% CI 1.38 to 1.42, respectively). Finally, precarious employment was associated with a 50% higher mortality rate among women (RR 1.50, 95% CI 1.48 to 1.53) and a 60% higher mortality rate among men (RR 1.60, 95% CI 1.58 to 1.62).

Covariate-adjusted associations between the employment quality typology and cause-specific mortality due to cancer, cardiovascular disease and unintentional injuries are presented in table 4. Results were generally consistent with those obtained in the analyses of all-cause mortality. Compared with the preceding analyses, however, estimated associations were somewhat smaller in magnitude for cancer and cardiovascular mortality and somewhat larger in magnitude for unintentional injury mortality. This was only true on the relative scale, since cancer and cardiovascular mortality account for a much greater proportion of deaths overall, resulting in larger absolute differences in cancer and cardiovascular mortality rates across employment quality groups. Furthermore, in a notable departure from the preceding analysis of all-cause mortality, portfolio employment was associated with a higher rate of cardiovascular mortality (among women) and unintentional injury mortality (among women and men) but not a higher rate of cancer mortality.

Sensitivity analyses stratifying the sample by age are presented in online supplemental appendix C (for all-cause mortality) and online supplemental appendix D (for cause-specific mortality). Results were generally consistent across younger and older workers. In some cases, however, associations were stronger among younger men (for all-cause mortality) and younger women (for both all-cause and unintentional injury mortality), relative to older workers in the sample. For example, precarious employment was associated with 60% and 83% higher rates of all-cause mortality among younger women and men (RR 1.60, 95% CI 1.47 to 1.75 and RR 1.83, 95% CI 1.70 to 1.86, respectively), whereas it was only associated with 47% and 52% higher rates of all-cause mortality among older women and men (RR 1.47, 95% CI 1.39 to 1.54 and RR 1.52, 95% CI 1.46 to 1.58, respectively). In another notable example of age-related patterning, portfolio employment was associated with a higher rate of all-cause mortality among younger women (RR 1.23, 95% CI 1.07 to 1.41), but not older women (RR 1.05; 95% CI 0.98 to 1.13). We did not observe meaningful differences between younger and older workers in the magnitude of associations between employment quality and either cancer or cardiovascular mortality.

Sensitivity analyses excluding respondents who died within the first 5 years of follow-up produced similar results.

DISCUSSION

This study examined the relationship between employment quality and mortality risk in the Canadian working population. Using an empirical typology capturing the variable nature and character of contemporary employment arrangements, we identified a clear gradient in all-cause and cause-specific mortality along the spectrum of employment quality. Workers in standard (ie, secure and gainful) and portfolio (ie, demanding but gainful) employment experienced the lowest mortality rates. They were followed by workers in marginal (ie, limited hours and earnings) and intermittent (ie, unstable and sporadic) employment. Finally, workers in precarious (ie, insecure and low-paying) employment experienced the highest mortality rates. These findings reinforce the need to move beyond a binary view of jobs as either 'standard' or 'precarious', encouraging a more nuanced understanding of contemporary employment arrangements and their health-related consequences.

Our results are consistent with prior research linking poor employment quality—and precarious employment in particular—to adverse health outcomes, including poorer general and mental health.[9–15 23 24 27 29] The current study extends that literature by focusing on mortality as a more definite and clinically significant outcome. Our findings also highlight the importance of distinguishing between different types of non-standard employment, including its high-quality and low-quality variations.[16 21 30] They also challenge the tendency in the literature to treat all non-standard jobs as necessarily precarious and therefore harmful to health. For instance, workers in portfolio employment—defined by a combination of highly demanding but economically rewarding employment conditions—experienced generally similar mortality rates to their standard counterparts, suggesting that there are ways of deviating from the 'Standard Employment Relationship' that do not necessarily present a risk to health. This aspect of our findings aligns with

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Table 4 Estimates from adjusted Poisson regression models describing the relationship between employment quality and cause-specific mortality, by sex/gender: 2006 Canadian Census Health and Environment Cohort

	Deaths	ASMR (95% CI)	RR* (95% CI)
Cancer mortality			
Women			
Standard employment	47 720	170.5 (168.7 to 172.3)	Reference
Portfolio employment	8595	161.8 (157.8 to 165.8)	1.01 (0.93 to 1.09)
Marginal employment	20 345	177.8 (175.3 to 180.4)	1.14 (1.08 to 1.21)
Intermittent employment	16 520	184.4 (181.2 to 187.5)	1.21 (1.12 to 1.32)
Precarious employment	11 230	197.1 (193.2 to 200.9)	1.28 (1.20 to 1.35)
Men			
Standard employment	63 275	194.2 (192.6 to 195.8)	Reference
Portfolio employment	27 835	176.7 (174.4 to 178.9)	0.95 (0.91 to 1.00)
Marginal employment	13 600	221.4 (217.6 to 225.2)	1.29 (1.21 to 1.38)
Intermittent employment	21 550	233.1 (229.9 to 236.3)	1.36 (1.27 to 1.45)
Precarious employment	11 665	256.4 (251.6 to 261.1)	1.40 (1.32 to 1.48)
Cardiovascular mortality			
Women			
Standard employment	9880	38.0 (37.1 to 38.9)	Reference
Portfolio employment	2045	41.4 (39.3 to 43.5)	1.23 (1.04 to 1.45)
Marginal employment	4760	43.2 (41.9 to 44.5)	1.31 (1.16 to 1.48)
Intermittent employment	3845	44.9 (43.3 to 46.6)	1.28 (1.11 to 1.53)
Precarious employment	3120	56.6 (54.5 to 58.8)	1.70 (1.51 to 1.91)
Men			
Standard employment	34 840	106.6 (105.3 to 107.9)	Reference
Portfolio employment	15 875	98.8 (97.1 to 100.5)	0.97 (0.91 to 1.03)
Marginal employment	8230	136.1 (133.1 to 139.1)	1.46 (1.34 to 1.60)
Intermittent employment	13 095	137.6 (135.1 to 140.1)	1.37 (1.25 to 1.49)
Precarious employment	7170	156.0 (152.3 to 159.8)	1.46 (1.36 to 1.57)
Unintentional injury mortality			
Women			
Standard employment	2520	9.0 (8.5 to 9.4)	Reference
Portfolio employment	515	11.4 (9.9 to 12.9)	1.33 (0.97 to 1.82)
Marginal employment	1420	11.4 (10.7 to 12.0)	1.44 (1.14 to 1.83)
Intermittent employment	1335	11.8 (11.0 to 12.6)	1.91 (1.41 to 2.58)
Precarious employment	1415	17.0 (15.9 to 18.0)	2.23 (1.82 to 2.73)
Men			
Standard employment	8835	26.2 (25.5 to 26.9)	Reference
Portfolio employment	5065	34.9 (33.5 to 36.3)	1.31 (1.16 to 1.48)
Marginal employment	2490	36.6 (35.0 to 38.1)	1.26 (1.06 to 1.50)
Intermittent employment	5060	40.2 (39.0 to 41.5)	1.80 (1.55 to 2.10)
Precarious employment	3965	53.5 (51.4 to 55.5)	2.00 (1.78 to 2.26)

*Adjusted for age, race/ethnicity, immigrant status, marital status, region of residence, rurality, education and household size.
ASMR, age-standardised mortality rate per 10 000 persons; RR, rate ratio.

labour market segmentation theory, which posits that the labour market is divided into distinct segments representing varying degrees and forms of labour market attachment.[22] Recent iterations of this theory emphasise the need to move beyond a binary view of 'good' and 'bad' jobs, as the increasing diversity of employment arrangements requires a more nuanced view of the labour market.[4 5] In the empirical typology presented here, for example, most non-standard jobs were found to sit somewhere between standard and precarious employment, with variable consequences for mortality. Collapsing these heterogeneous employment arrangements into a single 'precarious' category—a common practice in the literature—may lead to erroneous conclusions about non-standard employment and underestimate the true extent of health inequalities associated with precarious employment.

Our analyses of cause-specific mortality revealed some important nuances. While the overall gradient observed for all-cause mortality was generally maintained across cancer, cardiovascular and unintentional injury deaths, the magnitude of associations varied by cause. Associations were weaker for cancer and cardiovascular mortality and strongest for unintentional injury mortality. This pattern may reflect different underlying mechanisms linking employment quality to specific health outcomes. The stronger associations with unintentional injury mortality could be explained by the concentration of precarious workers in more hazardous occupations, as well as potential psychosocial factors such as stress and fatigue that may increase susceptibility to accidents.[15 24 31] The more modest associations with cancer and cardiovascular mortality might instead reflect the longer latency period required for chronic illnesses to develop,

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which may dilute associations, particularly when the exposure is measured at a single time point.[32] These cause-specific findings underscore the multiple potential pathways through which employment quality can affect health and highlight the particular vulnerability of precariously employed workers to preventable causes of death.

We also found some evidence of sex/gender and age differences in the association between employment quality and mortality. In general, we observed larger absolute and relative mortality inequalities among men relative to women, particularly in the case of all-cause and cardiovascular mortality. This finding is consistent with previous research on the employment-health relationship, which has shown that the health effects of non-standard employment and unemployment tend to be more pronounced among men—an effect that is often attributed to gendered norms concerning the 'male breadwinner' role.[16 33 34] Differences according to age were also evident, although to a lesser extent than sex/gender. While results were generally consistent among younger and older workers, relative inequalities in all-cause mortality and (among women) unintentional injury mortality were in some instances larger among younger workers relative to older workers. These instances of stronger associations among younger workers are likely to be a statistical artefact of their lower baseline rate of mortality, since relative inequalities tend to be larger in populations where absolute mortality rates are low.[35] Alternatively, they might reflect greater misclassification error among older workers, for whom a single, static measure of employment quality may be a relatively weaker proxy of lifetime histories of exposure.[36]

One particularly notable example of heterogeneity at the intersection of sex/gender and age—notable in the sense that it departs from the sex/gender patterning overall—was the finding that portfolio employment predicted an increased risk of all-cause mortality among younger women but no other group. An intuitive explanation for this finding could be that younger women face a uniquely challenging set of competing work and family demands.[37] It may be that the long hours associated with portfolio employment further frustrate those competing demands, with adverse implications for the health of younger women in the workforce.[38] This represents an important avenue for future research on sex/gender, employment quality and health.

Strengths of our study include the use of a large and nationally representative census cohort with >13 years of follow-up. Completion of the Canadian Census was mandatory, and one in five households was required to complete the long-form version. Thus, the 2006 CanCHEC represents one of the most powerful and reliable sources of information on the working population in Canada. With >13 years of follow-up, we were also sufficiently powered to jointly assess sex/gender and age differences in the relationship between employment quality and cause-specific mortality. This long follow-up period, allowing for the reliable estimation of sex-specific, age-specific and cause-specific mortality rates, represents an important advantage over more recent cycles of linked census and mortality data. Finally, we operationalised employment quality as a multidimensional construct—an approach that has advantages over simplistic and one-dimensional measures of employment quality that are prone to misclassification.

Our findings should also be interpreted in light of the following limitations. First, we lacked repeated measures of employment quality, limiting our ability to assess changes in employment conditions over time.[9] At the same time, previous research suggests that employment quality tends to be stable over the working life course, with relatively little evidence of mobility across segments of the labour market.[36 39] Second, our employment quality typology was constructed based on data collected nearly 20 years ago. The labour market may have undergone significant changes during that time. Nevertheless, evidence shows that the most substantial changes in the Canadian labour market occurred during the 1990s and 2000s, and these changes would be reflected in our empirical typology.[2] Additionally, while our typology captured multiple dimensions of employment quality (ie, hours, stability and earnings), other dimensions were missing due to a lack of measures on factors such as power relations (eg, union membership) and social protection (eg, pension coverage). However, given that adverse employment conditions tend to 'move together' in the labour market,[4 5] we believe that the inclusion of additional dimensions would have yielded a similar set of groupings. Finally, because baseline health status is not assessed in the census, we could not rule out unmeasured confounding due to health selection (ie, pre-existing health problems affecting both baseline employment quality and subsequent mortality risk).[35]

Taken together, the study findings point to substantial mortality inequalities along the spectrum of employment quality, with workers in precarious employment facing the highest risks, followed by those in other diverse types of non-standard employment. Policy interventions that promote access to high-quality jobs and protect workers exposed to precarious employment may yield substantial improvements in population health, including increased longevity.[40]

Contributors FVS developed the study idea. ATA performed the analysis with assistance from FVS. All authors discussed and interpreted the study findings. FVS wrote the first draft of the manuscript. All authors provided substantive comments and suggestions. All authors have reviewed the final version of the manuscript and have approved it for submission. FVS is the guarantor of the work.

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Ethics approval Data for this study were accessed through the Statistics Canada Research Data Centre Programme, which is subject to strict confidentiality and disclosure protocols outlined in the Statistics Act of Canada. These data are exempt from formal ethics review and approval, as stipulated in Article 2.2(a) of the Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans. All research outputs were vetted by Statistics Canada before release to ensure privacy.

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ORCID iDs
Faraz V Shahidi https://orcid.org/0000-0002-2789-1515
Anne E Fuller https://orcid.org/0000-0001-5930-5317
Nancy Carnide https://orcid.org/0000-0001-7892-5626
Roman Pabayo https://orcid.org/0000-0003-4018-4898
Brendan Smith https://orcid.org/0000-0003-2785-1246
Arjumand Siddiqi https://orcid.org/0000-0002-2458-9011
Peter Smith https://orcid.org/0000-0001-8286-4563

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